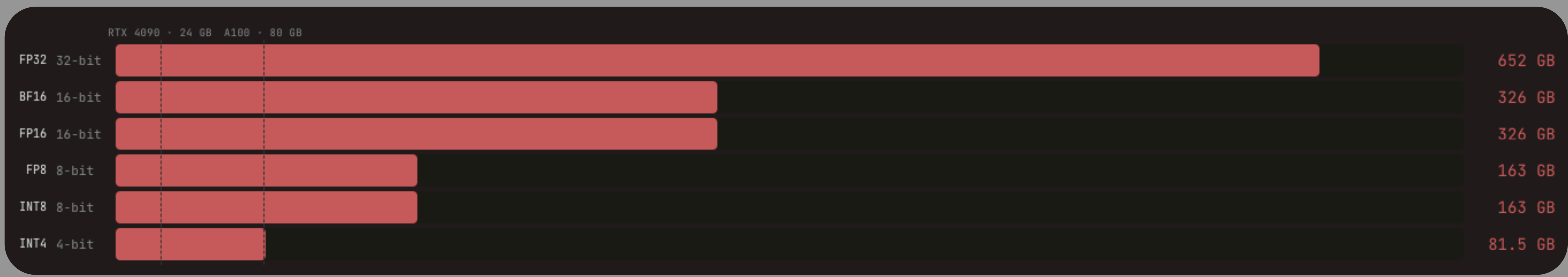
→ The cost of Bits

⇒ VRAM weight

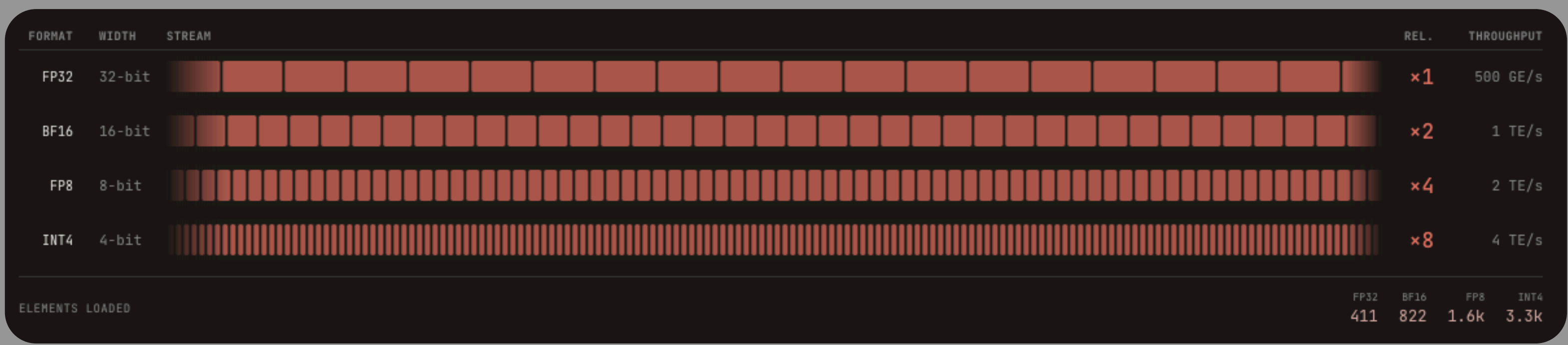
> understanding number formats and why precision matters for GPU performance

↳ Precision determines whether your model fits on one GPU, four, or not at all



↳ GPT-3 ; 175 parameters

⇒ Bandwidth Dividend



→ halving precision doubles effective memory bandwidth
No algo. changes, just smaller numbers thru the same pipe

→ Format Limits

⇒ The nearest fit

0.15

Try 0.2, 1.7, or 65500

FORMAT	KEPT AS	LAYOUT	GAP
FP32	0.150000006		close enough
TF32	0.150024414		close enough
BF16	0.150390625		close enough
FP16	0.150024414		close enough
FP8 E5M2	0.15625		close enough
FP8 E4M3	0.15625		close enough
INT8	0		1.50e-1
INT4	0		1.50e-1

sign · exponent · mantissa · integer · exact · close enough · large gap

> each format keeps the nearest number it can represent.
fewer bits means a bigger gap
between what you type and
what gets stored.

=> what breaks

INPUT
0.83
-1.27
0.45
2.91
-0.62

matmul → matmul → softmax

	FP32 ground truth	BF16 16-bit	FP8 e4m3
A	0.142	0.141602	0.140625
B	0.408	0.408203	0.40625
C	0.029	0.0290527	0.0292969
D	0.285	0.285156	0.28125
E	0.092	0.0917969	0.09375
	→ class B	→ class B	→ class B
		±0.000	±0.004

	FP32 ground truth	BF16 16-bit	FP8 e4m3
A	0.305	0.304688	0.3125
B	0.298	0.298828	0.3125
C	0.291	0.291016	0.28125
D	0.312	0.3125	0.3125
E	0.088	0.0878906	0.0859375
	→ class D	→ class D	→ class A
		±0.001	±0.015

	FP32 ground truth	BF16 16-bit	FP8 e4m3
A	0.305	0.304688	0.3125
B	0.298	0.298828	0.3125
C	0.291	0.291016	0.28125
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	→ class D	→ class D	→ class A
		±0.001	±0.015

> Small rounding errors add up
layer by layer
A low-precision format can look
fine early,
then pick the wrong answer at the end